ARCHITECTURE BLUEPRINT • WAMPSERVER & REACT NATIVE CLI

SoundWave: Free Spotify-Class Audio Engine

Technical System Design for React Native CLI, NativeWind (Tailwind CSS), Node.js, and WampServer MySQL.

AUDIO STORAGE & CLOUD COST

$0.00 / month

Zero MP3 hosting overhead via dynamic backend stream extraction

TARGET MOBILE STACK

React Native CLI + Tailwind

NativeWind v4 styling, react-native-track-player, and WampServer DB

1. Distributed System Topology



2. Architectural Foundations

This design replaces Flutter with **React Native CLI** styled with **Tailwind CSS (NativeWind)**, connected to a local **WampServer MySQL** engine and a Node.js intermediary server. Rather than running CPU-heavy scrapers directly in the mobile JavaScript runtime, stream resolution is handled cleanly by the local Node.js bridge using `@distube/ytdl-core`, returning ephemeral audio stream URLs directly to `react-native-track-player`.

> **WampServer Connection Strategy**
> WampServer runs MySQL on `localhost:3306` (default credentials user: `root`, password: *blank* or custom). While Node.js communicates directly via `localhost`, the Android emulator reaches your Node service at `http://10.0.2.2:3000` and a physical test phone reaches it via your local Wi-Fi IP (e.g., `192.168.x.x:3000`).

3. WampServer MySQL Database Schema & Node Connector

Import this DDL directly into **phpMyAdmin** in WampServer. Ensure your WampServer icon is Green and MySQL port 3306 is running.

```
      CREATE DATABASE IF NOT EXISTS soundwave_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE soundwave_db; CREATE TABLE users ( id INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50) NOT
NULL UNIQUE, email VARCHAR(100) NOT NULL UNIQUE, password_hash VARCHAR(255) NOT NULL, created_at TI
MESTAMP DEFAULT CURRENT_TIMESTAMP ) ENGINE=InnoDB; CREATE TABLE songs_cache ( id VARCHAR(120) PRIMA
RY KEY, -- YouTube Video ID or Audius Track ID title VARCHAR(255) NOT NULL, artist VARCHAR(255) NOT
NULL, cover_art_url TEXT, duration_seconds INT NOT NULL DEFAULT 0, created_at TIMESTAMP DEFAULT CUR
RENT_TIMESTAMP, INDEX idx_artist (artist), INDEX idx_title (title) ) ENGINE=InnoDB; CREATE TABLE pl
aylists ( id INT AUTO_INCREMENT PRIMARY KEY, user_id INT NOT NULL, title VARCHAR(150) NOT NULL, cre
ated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE C
ASCADE ) ENGINE=InnoDB; CREATE TABLE playlist_tracks ( playlist_id INT NOT NULL, song_id VARCHAR(12
0) NOT NULL, added_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, PRIMARY KEY (playlist_id, song_id), FORE
IGN KEY (playlist_id) REFERENCES playlists(id) ON DELETE CASCADE, FOREIGN KEY (song_id) REFERENCES
songs_cache(id) ON DELETE CASCADE ) ENGINE=InnoDB; CREATE TABLE favorites ( user_id INT NOT NULL, s
ong_id VARCHAR(120) NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, PRIMARY KEY (user_id,
song_id), FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE, FOREIGN KEY (song_id) REFER
ENCES songs_cache(id) ON DELETE CASCADE ) ENGINE=InnoDB;
```

4. Technology Matrix: React Native CLI Stack

Layer	Selected Package	Responsibility & Zero-Cost Role
Mobile Core	React Native CLI (TypeScript)	Direct native control, background threads, and Google Antigravity support
Styling	NativeWind (Tailwind CSS v4)	Spotify-grade dark UI styling via utility classes (`bg-black`, `text-green-500`)
Audio Playback	`react-native-track-player`	Hardware-level lock-screen controls, notification scrubber, and background playback
Backend Resolver	`@distube/ytdl-core` + Express	Dynamic, server-side audio-only demuxing (extracts highest bitrate opus/m4a stream)
Database	MySQL on WampServer (`mysql2`)	Local relational database with connection pooling and phpMyAdmin management

5. Implementation: Node.js Stream Resolver & Tailwind UI

The Node backend handles audio stream resolution. The React Native component utilizes **Tailwind CSS** classes via NativeWind to render Spotify's signature dark aesthetic and Mini Player.

```
      // 1. Node.js Express Route (backend/server.js) const ytdl = require('@distube/ytdl-core'); a
pp.get('/api/stream/:videoId', async (req, res) => { try { const info = await ytdl.getInfo(req.para
ms.videoId); const format = ytdl.chooseFormat(info.formats, { quality: 'highestaudio', filter: 'aud
ioonly' }); res.json({ streamUrl: format.url, title: info.videoDetails.title, artist: info.videoDet
ails.author.name }); } catch (err) { res.status(500).json({ error: err.message }); } }); // 2. Reac
t Native Tailwind Mini Player Component (src/components/MiniPlayer.tsx) import React from 'react';
import { View, Text, TouchableOpacity, Image } from 'react-native'; import TrackPlayer, { useIsPlay
ing } from 'react-native-track-player'; export const MiniPlayer = ({ track }: { track: any }) => {
const { playing } = useIsPlaying(); if (!track) return null; return ( <View className="h-16 bg-neut
ral-900 border-t border-neutral-800 flex-row items-center px-4 justify-between"> <Image source={{ u
ri: track.artwork }} className="w-12 h-12 rounded-md" /> <View className="flex-1 ml-3"> <Text class
Name="text-white font-bold text-sm" numberOfLines={1}>{track.title}</Text> <Text className="text-ne
utral-400 text-xs">{track.artist}</Text> </View> <TouchableOpacity onPress={() => playing ? TrackPl
ayer.pause() : TrackPlayer.play()} className="p-2"> <Text className="text-emerald-500 font-bold tex
t-lg">{playing ? '⏸' : '▶'}</Text> </TouchableOpacity> </View> ); };
```

6. Antigravity & React Native CLI Execution Checklist

Phase	Task Description	Antigravity Command / Context
01: Scaffolding	Generate React Native CLI app	`npx @react-native-community/cli init SoundWave --template react-native-template-typescript`
02: Tailwind Setup	Install NativeWind & Tailwind CSS	Configure `tailwind.config.js` and `babel.config.js` with nativewind presets
03: Audio Player	Configure Background Media playback	Install `react-native-track-player` and register `service.js` in `index.js`
04: WampServer Bridge	Connect Node.js to MySQL port 3306	Configure `mysql2.createPool` with user `root`, password `''` (Wamp default)

> **Ready for Build Execution**
> Place this specification document inside your Google Antigravity project folder to begin scaffolding the React Native CLI frontend, Tailwind styling, and WampServer MySQL integration.